Exercise on Combinatorial Proof

Show by combinatorial proof that: $\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}$

In a combinatorial proof, you must prove that both sides count the same number in different ways.

LHS: $\sum_{k=1}^{n} k\binom{n}{k}$ could count the number of ways $\binom{n}{k}$ to select a committee of k members $1 \leq k \leq n$, and then choose one the k members to be the chairperson, i.e., the committee should contain at least a chairperson and any other number of members.

RHS: $n2^{n-1}$ counts the same number because there are n choices for the chairperson and 2^{n-1} choices for the rest of the committee (the rest of the committee could be any subset, including the empty set, that does not contain the chairperson).